USRP Studies

Week 2: Lectures 3-5

Date

Overview

For the rest of my notes, I have decided to take a different approach to summarizing lectures; I will instead focus on main ideas of fundamental principles and take a more personal, narrative approach to the content. By doing so, I get more time to think about my research, and am able to connect ideas better.

Also, I skipped videos 4a,b,c, and 5b,c. These are covered in my school course in the fall, so I don't want to 'waste' time.

3a. Birthday Paradox Asymptotics

Birthday 'paradox': What is the probability that in a room of n people, at least two share the same birthday? Let's say none are leaplings, so 365 possible days. The problem with directly counting the probability is that there are many cases you must consider for pairwise matches. Hence we use the complement:

$$P(\text{no collisions}) = 1 \cdot (1 - \tfrac{1}{365}) \cdot (1 - \tfrac{2}{365}) \cdot \ldots \cdot (1 - \tfrac{n-1}{365}) \tag{1}$$

Typically, we log a giant product and consider its sum for asymptotics, since products are hard to analyze intuitively.

Using $\ln(1-x) \approx -x$ for small x:

$$\ln \Pr(\text{no collisions}) = \sum_{i=1}^{n-1} \ln\left(1 - \frac{i}{365}\right) \approx -\sum_{i=1}^{n-1} \frac{i}{365} \approx -\frac{n^2}{2 \cdot 365}$$

So with N possible days, $\Pr(\text{no collisions}) \approx e^{-n^2/2N}$. This hits $1/2$ at $n \approx 1.18\sqrt{N}$, which is about 23 for $N = 365$.

Remark 2.1. The main idea is that collisions start at $\sqrt{N}$, not N. This is because there are about $n^2/2$ *pairs*, and each pair collides with probability $1/N$. The same fact powers hash collisions and birthday attacks.

3b. Factorial Asymptotics (Stirling)

$$\ln n! = \sum_{k=1}^{n} \ln k \approx \int_1^n \ln x \, dx \approx n \ln n - n$$

Theorem 2.1. *(Stirling's Formula)* $n! \sim \sqrt{2\pi n}\ \left(\frac{n}{e}\right)^n$.

Remark 2.2. The calculus proof given here was quite underwhelming. The proof that gives the intuition on the $\sqrt{2\pi}$ constant is defered because we need knowledge of the central limit theorem (lecture 4).

3c. Binomial Coefficient Asymptotics

How big $\binom{n}{k}$ is depends on how big k is compared to n.

- **Small k:** $\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \left(\frac{en}{k}\right)^k$.
- **$k = \alpha n$:** plugging Stirling in gives $\binom{n}{\alpha n} \approx 2^{H(\alpha)n}$, up to a polynomial factor, where $H(\alpha) = -\alpha \log_2 \alpha - (1-\alpha)\log_2(1-\alpha)$ is the binary entropy.
- **Middle:** $\binom{n}{n/2} \sim \frac{2^n}{\sqrt{\pi n/2}}$. So the middle coefficient is only a $1/\sqrt{n}$ fraction of all 2^n subsets.

Remark 2.3. Near the middle, $\binom{n}{n/2+d} \approx \binom{n}{n/2} e^{-2d^2/n}$, which is a bell curve of width $\sqrt{n}$.

5a. Markov and Chebyshev Inequalities

Markov & Chebyshev Inequalities (5a)

Intuition motivating this lec:

More you know about R.V. ⇒ better bounds.

Let $X \geq 0$ be a R.V.

① Only know mean.

(1) prop (Markov's inequality) $P(X \geq a) \leq \frac{EX}{a}$, $a > 0$.

pf For $a > 0$, let

$$I = \begin{cases} 1, & X \geq a \\ 0 & \text{otherwise.} \end{cases}$$

Since $X \geq 0$, $I \leq \frac{X}{a}$. Taking expectations,

$$EI \leq E\left[\frac{X}{a}\right] \quad \therefore P(X \geq a) \leq \frac{EX}{a}. \quad \square$$

In the vid, we substitute $a = tEX$. So a is a multiple of the mean. I think the goal here is to make a 'unit-free'. It's also cooler, since you can say things like: "The median can never be twice the mean".

An interesting part of the pf in the video (not here) is setting $EX = 1$ (WLOG).

(2) prop (Chebyshev's inequality) If X is a R.V. with finite mean μ and variance σ^2,

$$\forall k > 0, \quad P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

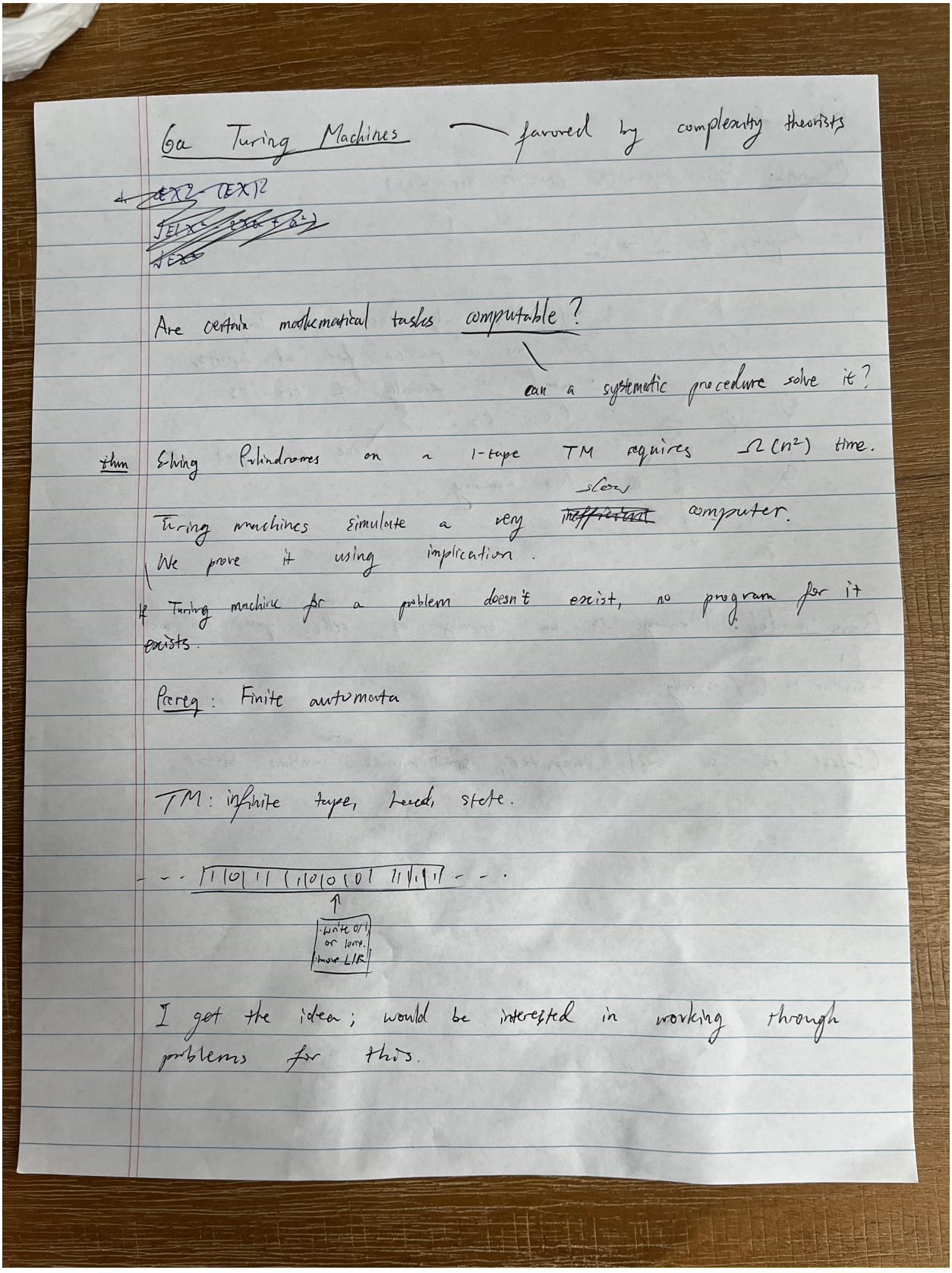
pf Apply Markov's inequality w. $a = k^2$, and R.V. $(X-\mu)^2$:

$$P((X-\mu)^2 \geq k^2) \leq \frac{E(X-\mu)^2}{k^2}$$

$$\Leftrightarrow P(|X-\mu| \geq k) \leq \frac{\sigma^2}{k^2}. \quad \square$$

② Know σ

6a. Turing Machines

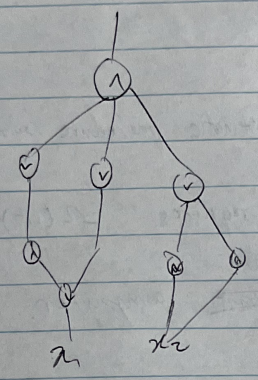


6b,c. Circuits and RAM Model

6b, c

Circuits: logic circuits (NAND / OR / ...)

computes $f: 2^n \to 2^n$.



- Handles only one input length.
- To solve a problem for all inputs, need a family of circuits $(C_1, C_2, \ldots)$ — one per input size.
- No memory

Ram model: Memory is an array of cells you can access.

similar to C / assembly.

Closest to a real computer, ~~some~~ it mimics ^ actual runtime better.